Lily Wang

Design Portfolio | *lily.jingyi.wang@gmail.com* | *linkedin.com/in/lily-jy-wang* | *github.com/lilywang899*

EDUCATION

University of Waterloo, *BASc in Computer Engineering* | GPA: 3.95 — Sept 2024 – May 2029

- President's Research Award (2025), Goulet Engineering Award (2025), President's Scholarship of Distinction (2024), Schulich Leader Scholarship Nominee (2024), Telesat Women in STEM Scholarship (2024)

SKILLS SUMMARY

Languages and Libraries: Python, PyTorch, C/C++, ROS, Matlab, TensorFlow, OpenCV, Pandas, NumPy, Scikit-learn

Software/Tools: Mujoco, Isaac Sim, Weights & Biases, HuggingFace, LeRobot, Rviz, FreeRTOS, Docker, Git

Hardware: Franka Panda, UR7e, Jetson Nano, Intel RealSense, Inspire Hand, Manus Glove, Meta Quest, IMU, LiDAR

WORK EXPERIENCE

Robotics Manipulation Research Intern, *Apptronik* — Apr 2026 – Aug 2026

- Built a world model based grasp-assistance teleoperation system for a **Franka arm** and 6-DoF **Inspire Hand**, which forecasts grasp failure from a single wrist camera, removing the need for multi-view cameras or costly haptic devices
- Trained a transformer latent-dynamics model on **DINOv2** visual features fused with pose, finger state, and per-finger force
- Implemented **CEM** planning over world model rollouts to select grasp-preserving action sequences
- Sped up world model training 2.66× via `torch.compile`, FlashAttention, fused LayerNorm and AdamW, and implemented KV caching for real-time hardware evaluation
- Implemented a learned **Hamilton–Jacobi** safety filter flagging object slip *before* failure in live hardware trials

Robotic Software Teleop Intern, *Apptronik* — Feb 2026 – Apr 2026

- Engineered a **ROS** data pipeline to record synchronized 30 Hz multimodal data with data augmentation and feature rebalancing to ensure the model relies on touch and proprioception, not just visual history
- Implemented neural network and inverse kinematics to train and fine tune robotic hand retargeting model
- Developed dexterity test bench by integrating **UR7e** with **Inspire hands**, using **Realsense T265** for real-time pose estimation
- Experimented with position and velocity-based arm controllers and rigorously tuned **IK solver** to deliver low latency, human-speed teleoperation

Machine Learning Researcher, *UWaterloo Software Engineering and System Engineering Lab* — Aug 2025 – Feb 2026

- Developed a multi-threshold, deep learning log anomaly detection system with **PyTorch** by implementing an end-to-end pipeline including tokenization, template extraction, and multi-resolution analysis at variable entropy thresholds
- Trained LSTM classifier to identify anomalous system behaviour using semantic representation of log templates, leveraging **GloVe** word embeddings, **HuggingFace** transformers to generate entropy signatures, and **NLTK** for stopword filtering

Data Engineer Intern, *Ford Motor Company* — May 2025 – Aug 2025

- Collaborated with cross-functional teams to design custom **parsing and feature-extraction** algorithms for diverse log structures, ensuring comprehensive coverage of potential anomalies
- Implemented an unsupervised deep learning model to parse and detect anomalies in global vehicle system logs, greatly streamlining defect detection across over 10 development teams

DESIGN TEAM

Autonomous Software Developer, *WATonomous (UWaterloo's Self-Driving Car Team)* — Jan 2025 – Apr 2025

- Developed real-time trajectory planning and control using **LiDAR and encoder data** for automatic emergency braking system
- Programmed publishers and subscribers using **C++, ROS2, Gazebo, and Foxglove** to visualize and monitor network data
- Based on open-source code, implemented a **nonlinear time-varying feedback controller** model using encoder data to guide differential drivetrain along pre-planned trajectory, improving accuracy over a pure PID controller

PROJECTS

Gripper | *C++, FreeRTOS, ESP32, PID controller, Solidworks* — Aug 2025 – Dec 2025

- Implemented custom gripper to fit robotic arm (*Dummy Robot*) using **Solidworks and Altium Designer**
- Developed gripper control by interfacing with **CAN** and **UART** peripherals using **FreeRTOS**

Perception |*C++, LibTorch, YOLO, OpenCV, ROS2* — Jun 2025 – Oct 2025

- Used Roboflow to label and train custom lightweight model, identifying objects manipulated by robotic arm
- Performed 3D object localization with trained model, **LibTorch and OpenCV** to guide robotic arm manipulation
- Implemented hand-eye calibration of robotic arm for an eye-in-hand camera configuration using **OpenCV, ROS2, and Rviz**